Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/13/2017
PI Name:	Khusid, Boris Ph.D.		
Project Title:	Kinetics of Electric Field-Driven Phase Transitions in Polarized Colloids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Newark	**State:**	NJ
Zip Code:	07102-1982	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	08/23/2013	**End Date:**	08/22/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 8/22/2019 per NSSC information (Ed., 8/7/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nelson, Emily Ph.D. (NASA Glenn Research Center) Sheehan, Christopher M.S. (ZIN Technologies, Inc.) Loewen, Hartmut Ph.D. (University of Dusseldorf) Telo da Gama, Margarida Ph.D. (University of Lisbon) Yethiraj, Anand Ph.D. (Memorial University of Newfoundland)		
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Task Description:	Motivation: Colloidal assembly provides the capability for scalable manufacturing of structured materials and devices for traditional and emerging industries. Although equilibrium structures and thermodynamics of colloids exposed to a field have been well studied, the use of electrically assisted assembly is limited by a lack of understanding of the role of time and spatially varying fields in directing and controlling non-equilibrium processes. Microgravity offers an opportunity to study field-driven phenomena on non-buoyancy-matched colloids by removing masking gravity effects, such as particle sedimentation, convection, and jamming. Objectives: Design and conduct microgravity tests in the International Space Station Advanced Colloids Experiment facility in long timescales to elucidate fundamental microscopic mechanisms of non-equilibrium phenomena underlying the field-assisted assembly of interacting colloidal particles and suggest novel routes for creating functional materials. The emphasis will be on the field-driven evolution of the mesoscale phase transitions, electro-hydrodynamic instabilities, and nucleation and growth of crystalline structures. The phenomena selected are logical candidates to take advantage of microgravity because terrestrial observations have failed to provide a comprehensive physical understanding of the underlying physics. Methodology: Dielectrophoretic experiments will be carried out on well-characterized colloids under well-defined conditions in terrestrial and microgravity environments. We use electro-hydrodynamic methods and apparatuses for the control and manipulation of particles which were developed in our NASA-supported projects, including parabolic flight tests conducted in 2002 and 2013. Molecular dynamics simulations, coarse-grained and continuum models will be employed to study field-induced colloidal structures. Deliverables: Detailed understanding of electrically driven non-equilibrium phenomena in colloids, apparatuses for the field-assisted colloid assembly, experimental database, and predictive models for the field-induced structure. The research will be integrated with educational and outreach initiatives in space sciences and related technologies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research has the strong potential to reveal the fundamental microscopic mechanisms of electrically driven phenomena in a suspension of interacting particles and develop a physical explanation for puzzling observations under terrestrial conditions that are not predicted by current theories. The main focus is on the appearance of a cellular structure in a colloid exposed to a spatially uniform field and understanding of the nucleation and growth of crystalline structures in a high-gradient and mechanisms of electro-hydrodynamic instabilities of concentration fronts. The obtained results will facilitate the development of new technologies for the control and manipulation of colloidal processes. Space applications: Electrical technologies can be employed in space to mitigate the effects of reduced gravity by providing an electrical body force. Results of ACE-T-E experiments will be useful for the development of new methods for electro-hydrodynamic control, manipulation, and separation of multiphase fluid flows in various space applications, for example, thermal management systems. Earth applications: The research should bring the understanding of electrically driven colloidal processes to the level useful in the design of apparatuses and processes for a broad range of applications in materials engineering, life science, electronics, and chemical industry. This will stimulate further "spin-offs" of electrical technologies.
Task Progress:	Major Accomplishments: The study was conducted within the scope of the originally-proposed research plan. The New Jersey Institute of Technology (NJIT) team worked in closed collaboration with NASA scientists on preparing International Space Station (ISS) Advanced Colloids Experiment (ACE) experiments. The main focus was on the development of requirements for a high-voltage amplifier and an optical system. The NJIT team carried out ground-based experiments on neutrally buoyant particles to specify basic parameters for the design of the ACE-E cell geometry, electrode configuration, voltage/frequency regimes, and preparing suspensions needed for the ISS experiments. It was shown that periodic cellular structures are formed in the presence of electric fields. Moreover, both AC and DC fields are essential to the formation of these structures. Presented examples illustrate the influence of the AC field frequency and the DC field strength on structure formation. It is important that the presence of a DC electric current is required to form a cellular pattern as it does not form on a coated surface. Presented examples illustrate the formation kinetics of the cellular structure. Development of Requirements for ISS Experiments: The Principal Investigator (PI) participated in two teleconferences with NASA and ZIN Technologies personnel on the ACE-T-E sample cell design in the spring 2017. If the gap between electrodes in the ACE-E cell lies in the range 25-100 µm, it would be sufficient to use a small (50x100x26 mm size) and inexpensive (about $440) Falco Systems WMA02 high-voltage amplifier http://www.falco-systems.com/ . This amplifier runs from a single 24-30V DC, 120 mA power supply and can be battery powered, if required. It provides 20x amplification up to +175V and -175V with respect to ground and has a full-power bandwidth from DC to 13kHz or 50kHz, if required. Its internal high voltage generator is inductor-free and hence does not spread magnetic interference in the sensitive ISS equipment. Operational requirements: An ACE-cell for ISS experiments will be fully filled with a suspension in the Earth facility. As the particles will sediment in Earth’s gravity during storage and the hyper-gravity during the spacecraft launch, specimens should be homogenized prior to experiments. A homogenized sample will then be exposed to an electrical field by applying a desired voltage/frequency for a certain period of time. To quantify effects of electric fields on colloidal processes, experiments will be carried out on each sample at different voltages and frequencies. After exposure to an electric field, a sample should be homogenized prior to another test. For each suspension, 3 - 6 magnitudes of applied voltage and several frequencies at each voltage will be tested. Several tests at selected voltage/frequency regimes will be repeated to evaluate reproducibility of microgravity experiments. The voltage/frequency regime and duration of a particular microgravity experiment will be determined in real-time depending on the previous results as data sets are transmitted to Earth. These parameters cannot be predicted based on observations of the field-driven processes in non-buoyancy-matched suspensions under Earth’s gravity. NJIT team demonstrated that it is possible to homogenize a sample by employing an electro-hydrodynamic flow generated in the ACE-E cell by applying a low frequency electric field. It would allow conducting several microgravity tests on each sample. This technique was

	successfully tested in ground-based experiments. Planned Work for Next Year: 1. Continue working with NASA project scientists and ZIN Technologies personnel on the selection of a high-voltage amplifier, voltage/frequency regimes, ACE-E cell design, and suspension compositions for ISS tests. 2. Continue ground-based experiments needed to specify science requirements for experiments aboard the ISS.
Bibliography Type:	Description: (Last Updated: 02/06/2025)
Abstracts for Journals and Proceedings	Khusid B. "Electric Field-driven structuring in suspensions." Concurrent Session: Complex Fluids #1. Presented at the 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Elele E, Lei Q, Khusid B. "Electric field-driven structuring in suspensions." Presented at the 2016 AIChE Annual Meeting, San Francisco, CA, November 13-18, 2016. 2016 AIChE Annual Meeting, San Francisco, CA, November 13-18, 2016. Abstract 425347, https://aiche.confex.com/aiche/2016/webprogram/Paper464969.html , Nov-2016
Abstracts for Journals and Proceedings	Khusid B, Elele E, Lei Q. "Electric field driven mesoscale phase transition in polarized colloids." Presented at the 69th Annual Meeting of the APS Division of Fluid Dynamics, Portland, OR, November 20–22, 2016. 69th Annual Meeting of the APS Division of Fluid Dynamics, Portland, OR, November 20–22, 2016. Abstract: M26.00001, V. 61, No. 20, http://absimage.aps.org/image/DFD16/MWS_DFD16-2016-002294.pdf , Nov-2016